PHASE 1 – T1: State of the art survey

A structured literature review covering the state of the art in multi-sensor multi-target tracking, with emphasis on maritime and harbor applications.

1. TARGET TRACKING ARCHITECTURES

A target tracking architecture is the overall structural design that defines how sensor data is processed, and used to maintain tracks of moving object.

WHERE DOES FUSION HAPPEN?

Centralized Fusion

All sensors send their data to **one fusion center** (e.g., the vessel), where tracking and data association are done in one place.

Advantages: It is statistically more optimal as it uses all information and there is no risk of double/counting or correlation issues

Disadvantages: Higher communication load and computational burden for large systems.

Decentralized Fusion

Each sensor node does its **own local tracking**, and only the resulting tracks are sent to a higher-level fuser. Fusion is distributed across multiple nodes.

Advantages: Scales better in large sensor networks as computation is distributed and the fusion center is not overloaded

Disadvantages: Correlation between tracks is hard to model and there is a risk of double-counting the information

WHAT IS BEING FUSED?

Measurement-level Fusion

The fusion center combines **raw sensor measurements** (range, bearing, AIS position, etc.) directly inside a single EKF.

Track-to-track Fusion

The fusion center combines **tracks** (state estimates + covariances) produced independently by each sensor's local tracker.

Conclusion: A centralized, measurement-level fusion architecture should be chosen as it is the easiest way to reliably combine all sensor data (a land-based mm-wave radar, a land-based stereo camera, an AIS receiver, and a GNSS receiver). All measurements go to one place (vessel, which acts as the fusion center) where, they are together. This is possible as the system has only few sensors, close to each other and therefore the computation is light.

2. DATA ASSOCIATION ALGORITMS

In the harbor surveillance system, there is multiple sensors (radar, camera, AIS) producing multiple detections at every step of the time. The tracker must decide **which detection belongs to which boat**, even when targets are close together, when clutter appears, or when some sensors miss detections. This matching process is called **data association**, and different algorithms exist to perform it with varying levels of accuracy and computational cost.

Algorithm	Definition	Accuracy	Computational Cost
Nearest Neighbor (NN)	Chooses the **single closest detection** for each track (based on distance).	**Low–Medium** Works only when targets are far apart. Fails in clutter or crossings.	**Very low** Fastest method
Global Nearest Neighbor (GNN)	Finds the **best overall assignment** between all tracks and all detections at once (Hungarian algorithm).	**Medium–High** More stable than NN, fewer identity swaps.	**Low–Medium** Polynomial time, still fast for small systems.
Probabilistic Data Association (PDA/JPDA)	Uses **all gated detections** and weights them by probability instead of choosing one. JPDA handles multiple targets.	**High** Very robust in clutter and crossing trajectories.	**Medium–High** Grows quickly with number of targets.
Multi-Hypothesis Tracking (MHT)	Keeps **multiple possible association hypotheses** over time and gets rid of unlikely ones.	**Very High** Best performance in difficult environments.	**Very high** Expensive, complex, often overkill for small systems.

Conclusion: Given the scope of the project, the **Global Nearest Neighbor (GNN)** algorithm is the most appropriate choice. After analyzing all required scenarios, it is clear that GNN provides **sufficient performance**. The Nearest Neighbor (NN) method would fail in situations where two or more boats are close to each other (for example scenario D), because it makes decisions independently for each track and simply selects the closest detection, which leads to identity swaps. On the other hand, PDA and MHT, while potentially more accurate, are significantly more computationally expensive. GNN is much simpler to implement, does not require extensive probabilistic calculations, and fits well within the project.

3. MULTI-SENSOR EKF FUSION

The EKF receives measurements from **multiple sensors** (radar, camera, AIS), which do **not** arrive at the same time and they have **different noise levels**. Therefore the EKF needs to know how to **combine multiple measurements at one time step** and what to do when **measurements arrive late.**

Method	Definition	Advantage	Disadvantages
Sequential Update	Update EKF one sensor at a time as measurements arrive.	Simple, works with asynchronous sensors, easy to implement.	Slightly less optimal than joint update.
Joint Update	Combine all measurements at the same timestamp into one big update.	Most statistically optimal, all sensors fused together.	Requires synchronized sensors, more complex.
Out-of-sequence Measurement Handling	Incorporate measurements that arrive late (after EKF already moved forward).	Handles real-world delays, improves accuracy.	Complex, requires rewinding or approximations.

Conclusion: Since the system uses several sensors, the incoming data will be asynchronous. Therefore, a method is needed to handle measurements that arrive at different times. The most suitable choice for this project is the **sequential update**, because the EKF can update the state immediately whenever a new measurement arrives, without waiting for all sensors to report. This is important in real-world conditions where some sensors may be delayed or temporarily degraded (e.g., due to bad weather). The ideal solution in theory would be **out-of-sequence measurement handling**, as it can correctly incorporate delayed measurements and therefore improve accuracy. However, it is significantly more complex to implement and exceeds the practical scope of this assignment.

4. MARITIME-SPECIFIC CONSIDERATIONS

Maritime environments introduce several challenges that directly affect multi-sensor tracking performance.

Consideration	Definition	Impact on Tracking	Disadvantages
AIS Integration	AIS gives absolute positions and vessel identity, but only for AIS-equipped boats and at low update rates.	Reliable but slow updates; missing for small or non-AIS vessels.	Helps maintain identity, stabilizes long-range tracking, supports fusion with radar/camera.
Sea Clutter	Radar returns from waves and reflections create clutter, especially near the coast or in rough seas.	More false detections → harder data association and gating.	Requires robust gating, filtering, and algorithms like GNN/JPDA.
Harbour Multipath	Buildings, cranes, ships, and containers block or reflect signals. Targets may disappear temporarily.	Tracks may drop or jump; intermittent detections.	Track coasting, careful deletion logic, and AIS as a stable reference when available.

Conclusion: Maritime environments introduce several sensor-related challenges that directly affect multi-target tracking.

Importance of AIS: AIS provides reliable identity and absolute position information, but only for equipped vessels and at a low update rate, so it must be fused with radar and camera to maintain continuous tracking.

Importance of GNN: Marine radar is affected by sea clutter and false alarms, especially near the coast or in rough sea states, which increases the need for robust gating and data association.

Challenge in Harbour: Harbour structures such as cranes, buildings, and large vessels create multipath and occlusions, causing intermittent sensor dropouts. These challenges motivate the use of a centralized multi-sensor fusion architecture, sequential EKF updates, and a robust data association method such as GNN to maintain stable tracks in realistic harbor conditions.

5. DEEP LEARNING APPROACH TO MULTI-TRACKING

Classical methods to multi-target tracking rely on:

- Motion models (constant velocity, constant acceleration)
- Sensor models (radar, camera, AIS noise models)
- Probabilistic filtering (EKF)
- Explicit data association (GNN, JPDA, MHT)

Their main advantages are that they are easy to tune, transparent, and explainable. They require very little data and are well suited for small-scale harbor scenarios. However, they can struggle when targets exhibit highly complex or non-linear motion. Deep learning approaches can potentially improve multi-target tracking by, for example

- Better handling of occlusions by using the appearance features to re-identify vessels after they disappear behind obstacles
- Learning complex motion patterns directly from the data

Nonetheless, these methods require large labelled datasets, are computationally expensive, and are unnecessary for tracking only 3–6 targets in a harbor environment.

Example: Deep learning innovations in South Korean maritime navigation: Enhancing vessel trajectories prediction with AIS data.

https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0310385

FINAL CONCLUSIONS

Having all of the considerations in mind the project should implement the following:

- Centralized, Multi-level fusion tracking architecture
- Global Nearest Neighbor (GNN) Data Association Algorithm
- Sequential Update Multi-sensor